

2 hours examples

Problem 1 (19RMM1)

Bob may always win. Let $f(n)$ be the smallest integer k such that $\frac{n}{k}$ is a perfect square. Bob may make sure that f of the number on the board is constantly decreasing as follows when it is his turn until it becomes one.

When given a number n with $f(n) = m$, we write $n = mk^2$. If Bob subtracts k^2 from the number, it will become $(m - 1)k^2$ and f of the result will be at most $m - 1$. Note that $f(n^k) = f(n)$ for odd k as n^{k-1} is a square and $f(n^k) = 1$ for even k since n^k is already a square. Therefore, on Amy's turn, she cannot increase $f(n)$.

Eventually, $f(n) = 1$ for the number n on the board meaning n is a square. Bob may then subtract n from itself and be left with zero as desired.

Problem 2 (CAKE)

Let I be the incenter. We cut along the altitudes from I to AC, BC, AB . Since the resulting pieces have a line of symmetry, they can be refitted to the corresponding positions on the box.

4 clubs; 4 hours

Problem 3 (14RUS107)

We show that this is possible. Take $\alpha = \frac{-1+\sqrt{29}}{2}$ meaning that $\alpha^2 + \alpha = 7$. Starting with the desired number n to construct, we write,

$$n = a_0 \cdot 1 + a_1 \cdot \alpha + a_1 \cdot \alpha^2 + \dots$$

We may start with $a_0 = n, 0 = a_1 = a_2 = \dots$. Starting with $i = 0$ and continuing until $a_i < 7$ for all i , at each step, we may guarantee $a_i < 7$ by subtracting 7 from a_i and incrementing a_{i+1}, a_{i+2} by one. This will not impact the equality and will decrease the sum $\sum_{i>0} a_i$. Since $\sum_{i>0} a_i$ decreases, we will eventually finish with a construction as desired.

Problem 5 (Z9B9D5CB)

The answer is 33.

For the construction, let the red cells form parallel diagonals three horizontal spaces apart.

To prove that this is optimal, we claim that each row of 9 cells may contain at most 3 red cells. It suffices to show that no two red cells in the same row may have less than two uncolored cells between them.

FTSOC, let this be the case. If there are three consecutive cells in a row with two red cells already filled. Then the three cells directed above and three cells directly below these three cells must all be unfilled for otherwise we have a 2×3 rectangle (horizontally long) with at least 3 red cells. However, in this case, we have a 2×3 rectangle (vertically long) with exactly one red cell. (!)

10 clubs; 5.5 hours

Problem 6 (23EGMO3)

The answer is 2^{k-1} .

For the construction, we write out the pairs of words that are opposites to each other.

$$(AA \dots A, BB \dots B)$$

$$(AA \dots AB, BB \dots BA)$$

...

Case 1: We have $AA \dots A$ or $BB \dots B$ in the dictionary.

In this case, filling the entire grid with A or B would suffice.

For the remaining $2^k - 2$ possible words, we may pair each word with the word you get if you flip each character from A to B or vice versa. By Pigeonhole, there must be a pair where both words are in the dictionary. Let the words be w_A, w_B . Let the characters of w_A be $c_1, c_2, \dots, c_k$. For a row i of our grid, we fill it in with w_A if c_i is A or w_B if c_i is B .

Using this method, each row is clear in the dictionary as it is either w_A or w_B . For each column, we notice that column i is w_A if c_i is A or w_B if c_i is B as desired.

We now prove the bound by finding a construction of $2^{k-1} - 1$ words where no grid suffices. We may select every single possible word that starts with A except $AA \dots A$. FTSOC, if a grid existed, since each row is a word, the start of each row must be A . As a result, the first column is $AA \dots A$ which is not in the dictionary. (!)

Problem 7 (81PTNMA2)

We claim the answer is 9, achieved by writing out 1 through 64 from left to right, and then top to bottom.

For the bound, note that we may always find a chain of consecutive adjacent squares 7 connections connecting 1 and 64. By Pigeonhole, at least one of the connections along the way has difference at least 9.

18 clubs; 6.5 hours

Problem 9 (04RUS91)

Note that every color is used at least once so there is a pair of lattice points one unit apart of different colors. WLOG, let $(0, 0)$ be blue and $(1, 0)$ be green.

Claim: No point on $y = 0$ is red.

Proof: We first check the color of $(0, 1)$.

If $(0, 1)$ was red, then $(0, 1), (0, 0), (1, 0)$ is a triangle that doesn't work.

If $(0, 1)$ was green, then taking $(0, 0), (0, 1), (m, 0)$ for any point $(m, 0)$ on $y = 0$ means that the entire row is not red.

We now consider the case where $(0, 1)$ is blue. Taking right triangle $(0, 1), (1, 0), (m, m - 1)$ for any m , we get $(m, m - 1)$ must be either green or blue. If it were green, taking $(m, m - 1), (m, 0), (0, 0)$ means $(m, 0)$ is not red. If it were blue, taking $(m, m - 1), (m, 0), (1, 1)$ means $(m, 0)$ is not red as well.

Doing this for all m gets us the result. $\square$

For any point (x, y) in the graph, we take $(x, y), (x, 0), (0, 0)$ if $(x, 0)$ is green or $(x, y), (x, 0), (1, 0)$ if $(x, 0)$ is blue to get (x, y) must not be red. However, this would imply no points in the plane is red. (!)

Problem 11 (14RUS93)

We claim the answer is $n - 3$ for odd n and $n - 2$ for even n . Call diagonals that are intersected exactly once "good".

For the construction, let $A_1, \dots, A_n$ be our vertices. We start with $A_1, \dots, A_4$ and draw diagonals A_1A_3 and A_2A_4 .

We then subsequently do the same thing with quadrilateral $A_1A_{2k}A_{2k+1}A_{2k+2}$ for all $k \geq 2$ until we go out of bounds.

Case 1: There exists a pair of good diagonals that intersect.

We prove this by induction. The cases $n = 3, n = 4$ are trivially true. Let $P(n)$ be the answer for a given n . We also define $P(2) = 0$ for the degenerate n -gon. We mark these two diagonals in red. Note that since these two diagonals are good and already intersect each other, no other diagonals may cross the red. Note that, as a result, we may split our polygon into four regions,

$$P(n) = P(w) + P(x) + P(y) + P(z) + 2$$

[clarification needed here?]

For $w + x + y + z = n + 4; w, x, y, z \geq 2$.

$$\begin{aligned} &= (w - 2) + (x - 2) + (y - 2) + (z - 2) - (\text{\#of odd numbers among } w, x, y, z) + 2 \\ &= (n - 2) - (\text{\#of odd numbers among } w, x, y, z) \end{aligned}$$

If n is even, we may achieve $n - 2$ by having w, x, y, z each be even. If n is odd, at least one of w, x, y, z must be odd meaning we're bounded at $n - 3$ as desired.

Case 2: There exist no good diagonals that intersect.

If we erase all diagonals except the good diagonals, we'd get a set of diagonals that do not intersect. Using a similar induction, we may prove there are at most $n - 2$ for odd and $n - 3$ for even.

26 clubs; 8 hours

Problem 12 (BMCQ24)

We claim the answer is $\frac{1}{2}$ in that there is a bijection between formations that allow a path and formations that do not.

There is a path from A to B if and only if it is possible to travel from a vertex at the top of the screen to the bottom along black edges only. This same situation is the same as there are five starting places and 5 close-able edges to traverse before making it to the end.

Problem 13 (99TTSSA1)

The answer is yes, and we will show the construction. We first construct the model in 2D. The area above the surface be a $9800 \times \frac{11}{2}$ rectangle and the area below be a $9800 - 4901 - 4901$ isosceles triangle. This forms a convex pentagon.

We now stretch the model with thickness x for some large x . Note that the area of the rectangle would be $9800 \cdot \frac{11}{2}$ while the area of the triangle would be $9800 \cdot \frac{99}{2}$ using Pythagorean Theorem to get the altitude. This means 90% of the volume is always below water no matter x .

The surface area below the water level would be $\left(\frac{11}{2} + \frac{11}{2} + 9800\right)x + 9800 \cdot \frac{11}{2} \cdot 2$ while the area below the water level would be $(4901 + 4901)x + 9800 \cdot \frac{99}{2} \cdot 2$. Since $\frac{\frac{11}{2} + \frac{11}{2} + 9800}{4901 + 4901} > 1$, for a sufficiently large x , we may ensure the former is greater than the latter as desired.

Remark: In hindsight 99^2 was overkill.

34 clubs; 11 hours

Problem 15 (13RUS95)

For the sake of contradiction, let this be impossible.

Claim: There exists no vertex where both sets are not positive.

Proof: Let the vertex itself have value x . Let the sum of the $n - 1$ other vertices in each of the sets be a, b . We have $x + a, x + b \leq 0$. Let y be the number opposite to x that occurs in neither set. Note that $x + a + b + y > 0$ implies $b + y > 0$ and $a + y > 0$. Therefore, choosing y as the center would yield a vertex where both sets are positive. $\square$

We construct a graph of $2n$ vertices where there is a vertex for each of the numbers. For a given vertex, we color it red if the sum of it with the $n - 1$ vertices that follow it going clockwise is positive and blue otherwise.

By the hypothesis, pairs of vertices $n - 1$ spaces apart must not both be red. By the claim, they can't both be blue either. Therefore, these pairs must have different colors.

Claim 2: Pairs of vertices n apart also have different color.

Proof: Clearly, they can't both be blue for the sum to be blue. If they are both red, then we let the first elements of the two sets be a, c respectively. If $a \geq c$, then we may center at a and have a vertex where both sets are positive. If $c > a$, we center at c . $\square$

Let the vertices be $1, \dots, 2n$. WLOG let 1 be red so we have $n, n+1, n+2$ blue. Looking at $n+2$, we then get 1, 2, 3 are all red. We then look at 3 and induct to eventually get a contradiction.

Problem 16 (20USEMO2)

We claim that the answer is $4^{1010} - 3^{1010}$. As a matter of fact, we claim that, if Calvin and Hobbes played this game on subsets of $\{1, 2, \dots, 2n\}$, the answer is $4^n - 3^n$.

Hobbes can achieve $4^n - 3^n$ by picking all the subsets of $\{1, \dots, 2n\}$ which at least one of $\{1, 2n\}, \{2, 2n-1\}, \dots, \{n, n+1\}$ are its subset. During the game, Hobbes may ensure none of these are won by Calvin by choosing $(2n+1) - (\text{Calvin's selection})$, which is always possible since the numbers are paired up.

To show there are exactly $4^n - 3^n$ such subsets, we induct. The base case $n = 1$ is clearly true since there's only $\{1, 2\}$ that work. If we add $2n-1, 2n$ to the set, all the previous sets S still work as well as $S + \{2n-1\}, S + \{2n\}, S + \{2n-1, 2n\}$. All sets $\{2n-1, 2n\} +$ subsets of $\{1, \dots, 2n-2\}$ also work. The overlap would be $S + \{2n-1, 2n\}$. In total, we have,

$$4(4^{n-1} - 3^{n-1}) + 2^{2n-2} - (4^{n-1} - 3^{n-1}) = 4^n - 3^n$$

We now show that $4^n - 3^n$ is maximized. FTSOC, let $x > 4^n - 3^n$ be the maximum that Hobbes may achieve. In a sample game on these x subsets for some moves by Calvin $a_1, \dots, a_{2020}$, let Hobbes' responses be $b_1, \dots, b_{2020}$.

We show that there must be at least 3^n sets Hobbes does not pick for him to win. We do this with induction once again.

Base case: $n = 1$. Calvin may get $\{\emptyset\}, \{1\}, \{2\}$ on his first move always.

Inductive step: Let Calvin's first move be a and Hobbes' first move be b .

Note that starting now, Calvin and Hobbes will pick from the $2n-2$ numbers $\{1, \dots, 2n\} - \{a, b\}$ out of which there are 3^{n-1} sets S that work Calvin may guarantee. Each of these sets exclude a, b .

Clearly, for each of these S , Calvin may guarantee $S + \{a\}$ since he has already chosen $2n-1$. Each of these sets includes a and excludes b .

If Calvin had started with choosing b and Hobbes chose c in response, then Calvin has 3^{n-1} sets he may guarantee from $\{1, \dots, 2n\} - \{b, c\}$. Adding $\{b\}$ to each of these sets, we'd have 3^{n-1} sets that Calvin may guarantee that includes b .

We have $3 \cdot 3^{n-1} = 3^n$ distinct sets as desired.

46 clubs; 15 hours

Problem 18 (00TTFSA5)

Call a row or column “positive” (charged) if there are more white cells than black cells on it, “neutral” (uncharged) if it has the same amount, and “negative” (charged) otherwise.

Claim: There cannot be two charged rows of different signs.

Proof: FTSOC let there be, WLOG row 1 positive and row 2 negative. Of the white cells in row 1, the columns that each cell is in must be negative for it to attack more black cells than white cells. Therefore, more than half the columns are negative. However, we have row 2 negative meaning more than half the columns are positive. (!) $\square$

For the sake of contradiction, let there exists a row or column that is not neutral. WLOG let row 1 be positive. Note that no other rows may be negative meaning there are more white cells than black cells. However, for any white cell in row 1, the column that it is in must be negative, which implies all columns are non-positive, meaning there are more black cells than white cells as desired. (!)

Problem 19 (RUSNBOX)

Let $F(r)$ be the volume of the set of points at most r from any point on or inside the outer box. Let $F'(r)$ be the volume of the set of points at most r from any point on or inside the inner box. Note that $F(r) \geq F'(r)$ always since any point at most r from a point inside the inner box is at most r from a point inside the outer box.

We may calculate,

$$F(r) = \frac{4}{3}\pi r^3 + (a + b + c)\pi r^2 + (ab + ac + bc)r + abc$$

$$F'(r) = \frac{4}{3}\pi r^3 + (a' + b' + c')\pi r^2 + (a'b' + a'c' + b'c')r + a'b'c'$$

We have,

$$F(r) \geq F'(r) \Leftrightarrow \frac{F(r)}{F'(r)} \geq 1$$

$$\Leftrightarrow \frac{(a+b+c)\pi r^2 + (ab+ac+bc)r + abc}{(a'+b'+c')\pi r^2 + (a'b'+a'c'+b'c')r + a'b'c'} \geq 1$$

Taking r to infinity, the LHS approaches $\frac{a+b+c}{a'+b'+c'}$ which we are given is < 1 . (!)

60 clubs; 18.5 hours